

Manas Arora

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EDUCATION

Maharaja Surajmal Institute of Technology, Delhi
Bachelor of Technology in Computer Science and Engineering

Aug 2023 – July 2027
CGPA: 8.73/10

Dewan Public School, Meerut
Class XII
Class X

Mar 2023
Percentage: 91.2%
Percentage: 92.6%

PROJECTS

NexChat – Real-Time Messaging Platform

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- Engineered a full-stack real-time messaging application using the MERN stack and Socket.io, supporting one-to-one chat, live typing indicators, online/offline presence, and DB-backed read receipts with WhatsApp-style blue and grey tick delivery indicators
- Re-architected WebSocket event broadcasting from direct socket IDs to Socket.io chat rooms, ensuring read receipt delivery remains reliable across multiple tabs and socket reconnections — since socket IDs are ephemeral and connection-specific
- Optimized user experience with debounced global user search to minimize redundant API calls on keystrokes, and debounced typing indicator socket events to prevent socket flooding during rapid input
- Containerized the backend with Docker (Node 20 Alpine) and deployed via a hybrid Render architecture — Dockerized web service for backend, static CDN for React frontend — masking cold-start latency through a root-level auth pre-fetch behind a branded loading screen

YT Chat AI – RAG Platform

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- Architected a 3-service production RAG system — React frontend, Node.js orchestration layer, and FastAPI Python microservice — where Node handles auth and session management while Python owns the entire retrieval and generation pipeline
- Solved cold-start reliability on free-tier deployment by implementing non-blocking ingestion (immediate 201 response) with recursive frontend polling to detect vector DB readiness — eliminating timeout failures without paid infrastructure
- Integrated NVIDIA NIM API to serve GPT OSS 120B for LLM inference, choosing it over direct OpenAI for higher compute capacity on free-tier usage; used LangChain to orchestrate retrieval, context injection, and conversational memory
- Configured cross-origin JWT auth with HttpOnly cookies and SameSite: none + trust proxy to handle secure session persistence across separately deployed frontend and backend services on Render

Quirk – Social Media Platform

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- Built a MERN-based social platform with posting, likes, follow system, and activity feeds
- Structured the codebase as a Turborepo monorepo to separate frontend and backend into independent packages with isolated dependencies and parallel build pipelines
- Designed REST APIs with Axios-based frontend integration, implementing optimistic UI updates on like/follow actions to eliminate perceived latency

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, C++, Python
Backend:	Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, Socket.IO
Frontend:	React.js, Next.js, Redux Toolkit, React Router, Tailwind CSS
Databases:	MongoDB, PostgreSQL, Pinecone (Vector DB)
AI/ML:	LangChain, RAG, Vector Embeddings, Semantic Search, NVIDIA NIM API
DevOps & Tools:	Git, GitHub, Docker, Postman, Prisma ORM, Turborepo, NPM
Deployment:	Vercel, Render, MongoDB Atlas
Core Concepts:	Data Structures & Algorithms, DBMS, OOP, System Design (Basics)

ACHIEVEMENTS

- Secured **8.73 CGPA** at MSIT Delhi, consistently among the top performers in the CSE department
- Built and deployed a **production-grade 3-service RAG platform** integrating Node.js, FastAPI, LangChain, NVIDIA NIM, and Pinecone — independently, within 4 days
- Containerized and deployed a **full-stack real-time chat application** using Docker on Render, solving a multi-tab WebSocket concurrency issue through room-based Socket.io broadcasting